In the **Scientific Skills Exercise**, you can diagnose a mineral deficiency in leaves of an orange tree. Micronutrient shortages are less common than macronutrient shortages and tend to occur in certain geographic regions because of differences in soil composition. One way to confirm a diagnosis is to analyze the mineral content of the plant or soil. The amount of a micronutrient needed to correct a deficiency is usually small. For example, a zinc deficiency in fruit trees can usually be cured by hammering a few zinc nails into each tree trunk. Moderation is important because overdoses of a micronutrient or macronutrient can be detrimental or toxic. Too much nitrogen, for example, can lead to excessive vine growth in tomato plants at the expense of good fruit production.

Scientific Skills Exercise

Making Observations

What Mineral Deficiency Is This Plant Exhibiting? Plant growers often diagnose deficiencies in their crops by examining changes to the foliage, such as chlorosis (yellowing), death of some leaves, discoloring, mottling, scorching, or changes in size or texture. In this exercise, you will diagnose a mineral deficiency by observing a plant's leaves and applying what you have learned about symptoms from the text and Table 37.1.

Data The data for this exercise come from the photograph below of leaves on an orange tree exhibiting a mineral deficiency.

INTERPRET THE DATA

1. How do the young leaves differ in appearance from the older leaves?
2. What change do you observe in the leaves, and where in the leaves does it appear? List the three nutrients whose deficiencies give rise to this symptom. Based on the symptom's location, which one of these three nutrients can be ruled out, and why? What does the location suggest about the other two nutrients?
3. How would your hypothesis about the cause of this deficiency be influenced if tests showed that the soil was low in humus?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Global Climate Change and Food Quality

Since plant growth is generally enhanced by increases in atmospheric CO_2 and by warming temperatures, global food production, as measured by plant biomass, is expected to increase in response to global climate change in certain parts of the world. But food production must also be judged by its quality. Unfortunately, there are indications that plants today, compared with plants in preindustrial times, are not taking up nutrients sufficiently to keep pace with their increased fixation of CO_2 into carbohydrates. For example, a study of 43 crops found significant declines in protein, Ca, P, Fe, riboflavin, and ascorbic acid from 1950 to 1999. Although this decline in food quality might be caused by a shift toward higher-yielding crops, studies of wild plants suggest that global climate change is the culprit. For example, a study revealed that the protein content of goldenrod (*Solidago canadensis*) pollen has declined by one third since the Industrial Revolution, and the change closely correlates with the rise in CO_2 that has occurred. Declines in the qualities of pollen as a food source might play a role in the widespread decline of honeybees, important pollinators whose decreasing population threatens crop production.

CONCEPT CHECK 37.2

1. Are some essential elements more important than others? Explain.
2. **WHAT IF?** If an element increases the growth rate of a plant, can it be defined as an essential element?
3. **MAKE CONNECTIONS** Based on Figure 9.17, explain why hydroponically grown plants would grow much more slowly if they were not sufficiently aerated.

For suggested answers, see Appendix A.

CONCEPT 37.3

Plant nutrition often involves relationships with other organisms

To this point, we have looked at plants as exploiters of soil resources, but plants and soil have a two-way relationship. Dead plants provide much of the energy needed by soil-dwelling bacteria and fungi. Many of these organisms also benefit from sugar-rich secretions produced by living roots. Meanwhile, plants derive benefits from their associations with soil bacteria and fungi. As shown in **Figure 37.9**, mutually beneficial relationships across kingdoms and domains are not rare in nature. However, they are of particular importance to plants. We'll explore some important *mutualisms* between plants and soil bacteria and fungi, as well as some unusual, nonmutualistic forms of plant nutrition.